

Design Philosophy: "Wrong-but-confident is worse than honestly-empty." The framework prioritizes deterministic constraints, clear data lineage, and strict schema isolation over heuristics to protect downstream hiring logic.

1. PIPELINE ARCHITECTURE

Ingest → Extract → Normalize → Merge → Score → Project

- Ingestion & Parsing:** Synchronous batch ingest reading structured (Recruiter CSV rows) and unstructured (Resume files, plaintext notes) streams safely into memory.
- Targeted Extraction:** Maps raw attributes using schema-specific filters (CSV parsers) or deterministic regular expressions (text streams) into decoupled staging fields.
- Data Normalization:** Standardizes primitive types and values into consistent formats prior to executing identity-matching pipelines.
- Deduplication & Merge:** Resolves identities across multi-source collisions using strict keys, tracking field histories through atomic updates.
- Confidence Assessment:** Calculates individual profile reliability weights based on source verification history and structural completeness.

2. NORMALIZATION CONTRACTS

- Phone Numbers:** Formatted strictly to international E.164 standards (e.g., +14155552671) via country-prefix mapping lookups.
- Temporal Data:** Standardized to YYYY-MM. Fragmented text values (e.g., "May '24", "Current") are parsed using explicit pattern matrices.
- Locations:** Mapped to official ISO-3166 alpha-2 country keys. Input text names are cross-checked against a localized geographic database.
- Skills:** Standardized via character trim, lowercasing, and exact lookup dictionary mapping (e.g., "ReactJS" and "React.js" resolve to "react").

3. IDENTITY & MERGE POLICIES

- Identity Resolution:** Deduplication triggers via exact match on normalized emails or phones. Fallback handles a composite hash of name + location.
- Source Priority Precedence:** Overlapping fields resolve using a deterministic tier matrix: **ATS JSON** (1) > **Recruiter CSV** (2) > **Resume PDF** (3) > **Recruiter Notes** (4). Higher tiers overwrite lower tiers.
- Confidence Formula:** Computed via a weighted combination:

Score = [Σ(W_f × C_f) / ΣW_f] - P_{missing}

Where W_f represents field priority weight, C_f maps source trust, and P_{missing} applies a strict penalty for missing essential contact keys.

4. CONFIGURABLE PROJECTION LAYER

- Kernel Isolation:** The internal system processes and stores a complete, immutable canonical profile. The projection layer functions as a separate post-process output filter.
- Dynamic Pointer Pathing:** Evaluates runconfigs via index-aware notation keys. An entry like "from": "emails[0]" pulls from the root canonical array and maps cleanly to the target schema output key.
- Missing Field Configuration:** Dynamic handlers resolve missing fields based on the input runtime config parameters:
 - null: Appends key with explicit null value.
 - omit: Strips out the field key entirely from JSON.
 - error: Halts execution, raising a validation exception.
- Metadata Toggling:** Strips tracking payloads (provenance, overall confidence) automatically during final JSON mapping if disabled.

5. EDGE CASE REMEDIATION

- Shared Corporate Contact Info:** Candidates sharing corporate mainlines or generic office mail boxes (e.g., info@company.com). Mitigation: If name strings match below a Jaro-Winkler threshold of 0.65, auto-merging blocks. Profiles stay isolated, flagged as "shared-provenance".
- Open Employment Windows:** Timelines omitting explicit end dates or using active keywords ("Present", "Ongoing"). Mitigation: Automatically maps variables to the active processing date runtime context, adding an isolated confidence penalty.
- Corrupted Text Encodings:** Malformed characters or log noise within raw recruiter notes. Mitigation: Ingestion layers discard fragments failing localized string safety parameters, logging an execution warning and falling back to null.

6. STRATEGIC SCOPE REDUCTIONS

- Descoped: Semantic NLP Skill Graphs:** Real-time vector spaces or LLM checks for semantic matching are removed. Replaced with highly performant exact-match normalization lookup matrices.
- Descoped: Asynchronous Messaging:** Distributed task queues (Celery/Kafka) are out of scope. System relies on an optimized, in-memory synchronous execution model designed to process thousands of records efficiently.